

Uplink PRB Utilization Forecasting in 5G Networks: Data Preprocessing, Clustering, and Feature Engineering

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Abstract—This work addresses the problem of forecasting uplink PRB utilization in a live 5G radio access network using high-frequency cell-level measurements. The task involves predicting 3 days ahead at 15-minute resolution (288 steps) and producing probabilistic outputs in the form of quantiles (q10, q50, q90). The dataset contains 226 cells over a 60-day period with no missing values. This report focuses on data preprocessing, exploratory clustering of cells, and feature engineering strategies to support accurate multi-step quantile forecasting.

Index Terms—5G networks, time series forecasting, PRB utilization, quantile regression, clustering, feature engineering

I. INTRODUCTION

The objective is to forecast uplink Physical Resource Block (PRB) utilization in a 5G radio access network. The forecasting horizon is 3 days ahead with 15-minute granularity. The output must be probabilistic, consisting of lower (q10), median (q50), and upper (q90) quantiles.

The dataset contains measurements from 226 cells collected over a 60-day period. Each cell exhibits distinct traffic patterns influenced by user behavior, location type, and network load dynamics.

II. DATASET DESCRIPTION

The dataset includes the following variables:

- Cell identifier
- Timestamp (15-minute intervals)
- Uplink PRB utilization percentage
- Uplink throughput volume
- Active uplink RRC-connected users

The dataset contains 1,301,760 records, with full temporal coverage and no reported missing values. Each cell is expected to contain 5,760 time steps.

III. DATA PREPROCESSING

A. Data Integrity Checks

Although no missing values are reported, the following validations are required:

- Ensure full 15-minute time continuity per cell
- Verify uniqueness of (cell, timestamp) pairs
- Detect and remove duplicate records if present

B. Scaling and Normalization

Feature scaling is required due to differing variable ranges:

- PRB utilization: percentage-based bounded variable
- Throughput: high dynamic range variable
- Active users: count-based discrete variable

Robust scaling or standardization is recommended. Log transformation may be applied to throughput if skewness is high. Scaling must be fitted only on training data to prevent leakage.

C. Time Index Construction

Data must be sorted by cell and timestamp. A strict 15-minute grid should be enforced per cell to ensure consistent sequence modeling.

IV. CELL CLUSTERING EXPLORATION

A. Motivation

Cells exhibit heterogeneous behavior depending on geographical and operational conditions. Clustering allows identification of similar traffic patterns, enabling shared modeling strategies and improved generalization.

B. Clustering Approaches

1) *Feature-Based K-Means*: Each cell is represented using aggregated statistics such as mean load, variance, peak usage, and correlation metrics. K-Means provides a scalable baseline approach.

2) *Dynamic Time Warping Clustering*: DTW-based methods capture similarity in temporal patterns even when peak activity is shifted in time. This is useful for daily traffic curves but computationally expensive.

C. Recommended Strategy

A two-stage approach is suggested:

- Initial clustering using feature-based K-Means
- Validation using DTW-based similarity on representative sequences

D. Cluster Feature Set

Useful descriptors include:

- Mean and maximum PRB utilization
- Standard deviation of load
- Throughput statistics
- Correlation between users and load
- Daily and weekly variability patterns

V. FEATURE ENGINEERING

A. Time-Based Features

Cyclical encoding is used to represent periodic behavior:

- Hour of day encoding using sine and cosine transforms
- Day of week encoding using sine and cosine transforms

B. Lag Features

Temporal dependencies are captured using lagged values:

- Short-term lags (15 min, 30 min, 1 hour)
- Medium-term lags (2 hours, 1 day)

C. Rolling Statistics

Rolling windows help smooth noise and capture trends:

- Rolling mean and standard deviation
- Rolling min and max
- Windows: 1 hour, 4 hours, 1 day

D. Derived Features

Additional engineered features include:

- Throughput per active user ratio
- Rate of change in PRB utilization
- Interaction terms between traffic variables

VI. MODELING IMPLICATIONS

A global forecasting model across all cells is recommended, optionally enhanced with cluster identifiers or embeddings. The model should support multi-step forecasting over 288 future time steps and output quantiles using a quantile loss function.

VII. WORKFLOW SUMMARY

The recommended pipeline is:

- Data validation and cleaning
- Feature scaling and preprocessing
- Feature engineering (lags, rolling, time encoding)
- Clustering analysis of cells
- Model training using shared representation
- Evaluation against baseline per-cell models

VIII. CONCLUSION

The dataset provides a strong foundation for time series forecasting with full temporal coverage and multiple informative features. Effective preprocessing, feature engineering, and clustering are key to building a scalable and accurate forecasting system for 5G network load prediction.

TODO.